

# TUAN NGUYEN

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## OBJECTIVE

Ph.D. Candidate in Computer Science with experience building and optimizing **trustworthy, scalable AI/ML systems** across **federated learning, RAG, and GPU-accelerated deep learning**. Skilled in developing efficient training and inference pipelines, profiling bottlenecks, and translating research ideas into scalable implementations.

## EDUCATION

**University of Illinois Urbana-Champaign (UIUC) & VinUniversity** Jan 2023 – Present  
Ph.D. Candidate in Computer Science (Joint Program)

- Advisors: [Prof. Kok-Seng Wong](#), [Prof. Khoa Doan](#) (VinUniversity), and [Prof. Fan Lai](#) (UIUC)
- Concurrently pursuing M.S. in CS at UIUC (started Fall 2025) — Awarded Vingroup Scholarship

**VNU-University of Engineering and Technology (VNU-UET)** Aug 2015 – Jul 2019

Bachelor of Science in Computer Science — Graduated with High Distinction

- Ranked 1st in a class of 30 — Awarded VNU-UET Scholarship for Academic Excellence

## SELECTED PUBLICATIONS

A complete list of publications can be found on my [Google Scholar profile](#).

\* indicates equal contribution

- **Tuan Nguyen**, Khoa D Doan, Kok-Seng Wong. “FLAT: Latent-Driven Arbitrary-Target Backdoor Attacks in Federated Learning.” In arXiv, 2025. [link](#).
- **Tuan Nguyen**, Dung Thuy Nguyen, Khoa D Doan, Kok-Seng Wong. “Non-Cooperative Backdoor Attacks in Federated Learning: A New Threat Landscape.” In arXiv, 2024. [link](#).
- Ngoc-Hieu Nguyen, Tuan-Anh Nguyen, **Tuan Nguyen**, Vu Tien Hoang, Dung D Le, Kok-Seng Wong. “Towards Efficient Communication Federated Recommendation System via Low-rank Training.” In WWW, 2024. [link](#).
- Thuy Dung Nguyen\*, **Tuan Nguyen**\*, Phi Le Nguyen, *et al.* “Backdoor Attacks and Defenses in Federated Learning: Survey, Challenges and Future Research Directions.” *Eng. Appl. Artif. Intell.*, 2024. [link](#).
- Thuy Dung Nguyen, **Tuan Nguyen**, Anh Tran, Khoa D Doan, Kok-Seng Wong. “IBA: Towards Irreversible Backdoor Attacks in Federated Learning.” In NeurIPS, 2023. [link](#).
- **Tuan Nguyen**\*, Phi Nguyen\*, Dai Tran, Hung Pham, *et al.* “Ensemble Learning of Myocardial Displacements for Myocardial Infarction Detection in Echocardiography.” *Front. Cardiovasc. Med.*, 2023. [link](#).

## SKILLS

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| Programming Languages | Python, C/C++, Java, Kotlin, SQL                               |
| AI/ML Frameworks      | TensorFlow, PyTorch, Scikit-Learn, Hugging Face (Transformers) |
| GPU/Systems           | CUDA, Tensor Cores, Nsight Systems, Nsight Compute             |
| Tools & Platforms     | Git, Bash, Docker, Kubernetes, GitHub Actions (CI/CD), OpenCV  |

## EXPERIENCE

**Research Assistant**, University of Illinois Urbana-Champaign Aug 2025 - Present

- Developing and optimizing data compression techniques within Retrieval-Augmented Generation (RAG) systems, enhancing efficiency, reducing computational overhead, and improving latency for **LLMs**.
- Evaluating the impact of various compression strategies on RAG system performance, including accuracy, response quality, and resource utilization, ensuring robust and scalable solutions.

- Optimized CUDA-based convolution kernels for batched CNN workloads using fused implicit-GEMM, Tensor Core-accelerated matrix multiplication, and low-level GPU performance tuning.
- Analyzed execution bottlenecks with Nsight Systems and Nsight Compute, reducing convolution op time from **86.33 ms** to **15.42 ms** while preserving model accuracy at **0.8714**.

**Research Assistant**, College of Engineering & Computer Science (CECS), VinUniversity Jan 2023 - Present

- Conducted research on a project funded by VISHC, developing the [Privacy-Preserving, Robust, and Explainable Federated Learning Framework](#) for healthcare systems, enhancing data security and model reliability.
- Published multiple papers from this project, including a **survey on backdoor attacks in federated learning**, methods for **backdoor attacks in FL**, and an **empirical study on federated unlearning**.

**Research Assistant**, VinUni-Illinois Smart Health Center (VISHC), VinUniversity Jan 2022 - Dec 2022

- **Video-based AI for Cardiac Function Assessment:** Developed video-based AI models for cardiac function assessment and disease prediction, leveraging machine learning for improved medical diagnostics as part of a VinIF-funded project.
- Published in *Front. Cardiovasc. Med.*: “Ensemble Learning of Myocardial Displacements for Myocardial Infarction Detection in Echocardiography” from this work.

**Research Engineer**, UET-AILab, VNU-UET Jan 2019 - Dec 2021

- Led the development of a comprehensive AI system (funded by VinIF) for cardiac analysis, including algorithms for Left Ventricular Global Longitudinal Strain (LVGLS) measurement and Left-Ventricle detection with Ejection Fraction (EF) calculation, significantly enhancing diagnostic accuracy for medical imaging.
- Designed and implemented a Medical Image Data Storage System and an accompanying Mobile App for medical image interpretation, improving data accessibility and usability for clinicians.
- Co-authored a patent (National Office of Intellectual Property of Vietnam, 2022): “[Automated Quantification of Ejection Fraction directly from Echocardiography video](#).”

## ACADEMIC SERVICES

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- Reviewer for ICLR (2024, 2025, 2026), NeurIPS (2024, 2025, 2026), AISTATS (2026), and ICML (2025, 2026); NeurIPS '25 Datasets and Benchmarks track; and workshops: Backdoors (NeurIPS '23), DIG-BUG (ICML '25)

## TEACHING

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**Teaching Assistant**, VinUniversity Feb 2022 - June 2025

- Courses: Cybersecurity (Sp24), Intro to Programming (Fa22-Fa24), OOP & Data Structures (Sp23)

**Teaching Assistant**, VNU-University of Engineering and Technology Sep 2019 - Dec 2021

- Courses: Intro to Programming Languages (Fa19–Fa21), Advanced Programming (Sp20, Sp21)

## AWARDS AND ACHIEVEMENTS

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- Finalist & Demo Showcase Team (The Liemz), K2 Think Hackathon — K2 Tutor (2025).
- Vingroup Scholarship for M.S. in Computer Science at UIUC (2025).
- Full Scholarship for Ph.D. in Computer Science at VinUniversity (2023).
- Second place in the National Mathematics Contest for University Students (2019).
- Young Face Vietnam National University Award (2019).
- Fully-funded sponsorship for the KAIST CS Camp in South Korea (2019).
- VNU Mitsubishi Scholarship awarded by Tokyo-Mitsubishi UFJ Bank, Japan (2018).
- Second place in ACM ICPC Vietnam National Programming Contest (2017).
- Second place in the National Informatics Contest for High School Students (2015).